Cody A Dailey

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Education

2024	**PhD**	**University of Georgia (UGA)** Under the supervision of Justin Bahl, PhD Department of Epidemiology and Biostatistics Concentration in Data Analysis and Modeling Dissertation/Thesis Title: Spatial Epidemiology of Seasonal Influenza in the United States https://esploro.libs.uga.edu/esploro/outputs/9949694128302959
2017	**MPH**	**UGA** Department of Epidemiology and Biostatistics Dual Degree, Concentrations in Biostatistics and Epidemiology
2015	**BS**	**UGA** Franklin College of Arts and Science Biology, minor in French

Funding

2023-2024	Research Assistantship Principal Investigator: Justin Bahl, PhD Center for Applied Pathogen Epidemiology and Outbreak Control Pathogen Genomics Centers of Excellence
2021-2022	Research Assistantship Principal Investigator: Justin Bahl, PhD Department of Epidemiology and Biostatistics, UGA Funded via Centers for Disease Control and Prevention Contract No. 75D30121C11990
2021	Research Assistantship Principal Investigator: Andreas Handel, PhD Department of Epidemiology and Biostatistics, UGA
2020-2021	Research Assistantship Principal Investigator: Karen E Watkins, PhD Department of Lifelong Education, Administration & Policy, UGA Partnership among UGA, Geneva Learning Foundation, World Health Organization, and Bill & Melinda Gates Foundation
2016-2018 & 2020-2021	Teaching Assistantship Department of Epidemiology and Biostatistics, UGA
2018-2020	Interdisciplinary Disease Ecology Across Scales (IDEAS) National Science Foundation (NSF) Research Traineeship (NRT) Program Director: Vanessa Ezenwa, PhD Odum School of Ecology, UGA Grant No. DGE-1545433

Teaching Experience

2021,2022 **Teaching Assistant** – Simulation Modeling in Immunology, Summer Institute in Statistics and Modeling in Infectious Diseases, University of Washington;
Instructors: Andreas Handel and Paul Thomas

2021 **Instructor of Record** – Fundamentals of Epidemiology // EPID4070

2021 **Teaching Assistant** – Environmental and Occupational Epidemiology // EPID8070

2020 **Teaching Assistant** – Infectious Disease Epidemiology: A Model-based Approach // EPID8510E

2019 **Graduate Instructor & Tutor** – Introduction to Epidemiology I // EPID7010/E

2019 **Workshop Designer & Instructor** – SAS and R: A Programming Primer

2018-2019 **Undergraduate Co-Instructor** – Fundamentals of Epidemiology // EPID4070

2016-2018 **Teaching Assistant** – Fundamentals of Epidemiology // EPID4070

2017 **Teaching Assistant** – Introduction to Epidemiology I // EPID7010

Professional Experience

2025 **Researcher**; Applied Molecular Epidemiology Research Group; College of Veterinary Medicine; University of Georgia
Under the supervision of Justin Bahl

2020 **Intern**; French National Institute of Research for Sustainable Development (Institut de recherche pour le développement)
Under the supervision of research director Benjamin Roche in the MIVEGEC group (Maladies Infectieuses et Vecteurs: Ecologie, Génétique, Evolution et Contrôle)
Analyzed data from the PANIC (PAthogènes NIches et Contrôle) which aimed to describe the ecological niche of *Plasmodium falciparum* in Burkina Faso; will develop mechanistic models for the estimation of infection incidence and the effects of antimalarial interventions, particularly bed net usage and reminders, to the end of exploring optimal intervention strategies

2018-2019 **Collaborator** with Manager of Statistical Modeling and Analytics, Piedmont Healthcare

2017 **Intern**; Piedmont Athens Regional Medical Center
Under the supervision of the Quality Performance and Improvement Manager
Designed, applied, and evaluated predictive models for Emergency Room patient admission; explored burden and recommended intervention efforts for heart failure readmission in the development of a heart failure unit.

2016 **Intern**; Piedmont Athens Regional Medical Center
Under the supervision of the Infection Prevention Manager
Developed, implemented, and evaluated Safe Zone program (procedure for limited-entry to isolation patient rooms); constructed isolation precautions policies; updated FAQs and At-home Instructions for common afflictions.

Professional Training

2018-2020 IDEAS NRT Program
Odum School of Ecology, UGA
Emphasis on interdisciplinary, across scale work; coursework included: Responsible Conduct of Research // GRSC8550; Population Biology of Infectious Diseases // ECOL6150; Fundamentals of Disease Biology // ECOL8510; Interdisciplinary Problem-solving in Infectious Disease Ecology // ECOL8530

2019 Eukaryotic Pathogen Database Resources (EuPathDB) Workshop Participant, UGA

2019 Introduction to Scientific Programming Workshop Participant, IDEAS

2019 Modeling Infectious Diseases Workshop Participant, IDEAS

2019 Communicating Science to Journalists and News Media Workshop Participant, IDEAS

2018 Communicating Science to Policymakers Workshop Participant, IDEAS

2017 Modeling Immunology to Influenza Infection Summer School Trainee, Emory University

Publications

PUBLISHED ARTICLES

7. Megan E J Lott, William A Norfolk, Cody A Dailey, Amelia M Foley, Carolina Melendez-Declet, Megan J Robertson, Stephen L Rathbun, Erin K Lipp, Direct wastewater extraction as a simple and effective method for SARS-CoV-2 surveillance and COVID-19 community-level monitoring, FEMS Microbes, Volume 4, 2023, xtad004, https://doi.org/10.1093/femsmc/xtad004
6. Watkins, K.E., Sandmann, L.R., Dailey, C.A. et al. Accelerating problem-solving capacities of sub-national public health professionals: an evaluation of a digital immunization training intervention. BMC Health Serv Res 22, 736 (2022). https://doi.org/10.1186/s12913-022-08138-4
5. Mallis N, Dailey C, Drewry S, Howard N, Cordero JF, Welton M. SARS-CoV-2 infection and e-cigarette use, binge drinking, and other associated risk factors in a college population. Journal of American College Health. 2022 Mar 14:1-5.
4. Huber L, Giguère S, Hart KA, Slovis NM, Greiter ME, Dailey CA, Cohen ND. Association between antimicrobial treatment of subclinical pneumonia in foals and selection of macrolide-and rifampicin-resistant Rhodococcus equi strains at horse-breeding farms in central Kentucky. Journal of the American Veterinary Medical Association. 2021 Mar 15;258(6):648-53.
3. Álvarez-Narváez S, Giguère S, Berghaus LJ, Dailey C, Vázquez-Boland JA. Horizontal spread of Rhodococcus equi macrolide resistance plasmid pRErm46 across environmental actinobacteria. Applied and environmental microbiology. 2020 Apr 17;86(9):e00108-20.
2. Thompson JM, Eick SM, Dailey C, Dale AP, Mehta M, Nair A, Cordero JF, Welton M. Relationship between pregnancy-associated malaria and adverse pregnancy outcomes: a systematic review and meta-analysis. Journal of tropical pediatrics. 2020 Jun;66(3):327-38.
1. Huang K, Chen J, Xu R, Jiang X, Ma X, Jia M, Wang M, Huang J, Liao Q, Shan Z, Dailey C. Molecular evolution of hepatitis C virus in China: a nationwide study. Virology. 2018 Mar 1;516:210-8.

PREPRINTS/PENDING

- Cody Dailey, Christina Næsborg-Nielsen, Justin Bahl. Regional Commuting Patterns and Influenza-like Illness in the United States.
- Cody Dailey, Ludy Registre Carmola, Lambodhar Damodaran, Guppy Stott, Justin Bahl. Network-based Identification of Influenza Transmission Zones in the United States.
- Swan Tan, Lambodhar Damodaran, Cody Dailey, Jiani Chen, Rebecca Kondor, Leonard Moise, Anne S De Groot, Justin Bahl. Sequence-based evaluation of CD4+ T cell immune landscape of H3N2 seasonal influenza A virus.
- Lambodhar Damodaran, Cody Dailey, Swan Tan, Norman Hassell, Rebecca Kondor, Ludy Carmola, John M Drake, Pejman Rohani, Justin Bahl. Identifying source and sink regions of seasonal influenza A(H3N2) virus transmission in the United States, 2011-2020.
- Leke Lyu, Mandev Gill, Guppy Stott, Sachin Subedi, Cody Dailey, Gabriella Veytsel, Magdy Alabady, Kayo Fujimoto, Ryker Penn, Pamela Brown, Roger Sealy, Justin Bahl. Tracing SARS-CoV-2 Clusters Across Local Scales Using Genomic Data. medRxiv 2024.09.18.24313896; doi: https://doi.org/10.1101/2024.09.18.24313896
- Leke Lyu, Gabriella Veytsel, Guppy Stott, Spencer Fox, Cody Dailey, Lambodhar Damodaran, Kayo Fujimoto, Pamela Brown, Roger Sealy, Armand Brown, Magdy Alabady, Justin Bahl. Characterizing Spatial Epidemiology in a Heterogeneous Transmission Landscape Using a Novel Spatial Transmission Count Statistic. medRxiv 2023.12.28.23300535; doi: https://doi.org/10.1101/2023.12.28.23300535

Presentations

2. Dailey C and Bahl J. Genetic diversity of malaria according to surveillance strategy and study design. EPIDEMICS 7 / 7th International Conference on Infectious Disease Dynamics. Charleston, SC, USA. 6 Dec 2019.

1. Royster C, Dailey C, Parvathi A, Adams L, Rush L, Cai X, Zhang M. A 17-Year Overview of the Opioid Epidemic in Georgia. UGA Interdisciplinary Opioid Epidemic Symposium. 15 Oct 2018

Software

Contributor; Dynamical Systems Approach to Immune Response Modeling (DSAIRM) R Package.
Andreas Handel (2017). Learning infectious disease epidemiology in a modern framework. PLoS Computational Biology, 13(10), e1005642. URL https://doi.org/10.1371/journal.pcbi.1005642

Contributor; Dynamical Systems Approach to Infectious Disease Epidemiology (DSAIDE) R Package.
Andreas Handel (2020). A software package for immunologists to learn simulation modeling. BMC Immunology 21(1). URL https://doi.org/10.1186/s12865-019-0321-0

Mentoring

2019-2023 **PhD Student Mentor**; Practice, Research, and Mentorship in Epidemiology (PRIME) Working Group
Acted as group leader to undergraduates interested in research activities; facilitate research study design, implementation, analysis, and dissemination

Service

2024 **Secondary Instructor** for Nextstrain Narrative Training Workshop
Designed tutorial materials and guided participants through example data analyses and visualizations using Narrative documents

2019-2022 **Social and Service Chair**, Graduate Scholars of Epidemiology and Biostatistics (GSEB) Student Organization

Organized events including SAS and R: A Programming Primer Workshop, Holiday Cards for Hospice Patients, Charity 5K to fund CPH Food Pantry, Coffee for Cops Appreciation Effort

2019 **Instructor** of Infectious Diseases for Public Health Merit Badge, Boy Scouts of America
Designed and instructed lectures and active learning modules to give a broad basis of infectious disease biology and epidemiology

2017-2018 **Vice President** in Epidemiology, **Co-Founder**, GSEB
Founded student organization; organized events including Holiday Cards for Hospice Patients, Charity 5K to fund CPH Food Pantry, Coffee for Cops Appreciation Effort

Awards

2021 Graduate Service Award

2020 Ann & Tipton Golias International Travel Fund

Media

2020 College of Public Health Student Profile
https://publichealth.uga.edu/student-profile-cody-dailey/

Complementary Skills

Statistical Analysis Programming:
- R (proficient)
- SAS (proficient)

Languages:
- English (native)
- French (working knowledge; lu, écrit, parlé)

General Computing and Programming:
- High-performance Computing Clusters (limited)
- Linux (limited)
- Parallel Processing (familiar)

Analytical Techniques/Methodology
- Stratification Analysis
- Exploratory Data Analysis
- Data Visualization
- Data Management
- Adjustment
- Network Analysis
- Measures of Association: risk ratios, rate ratios, odds ratios, correlation
- Regression: linear, generalized linear, logistic, mixed-effects, non-linear (limited), longitudinal, generalized estimating equations
- Bayesian Computation: Metropolis-Hastings, Gibbs sampler (limited)
- Machine Learning: classification and regression trees, random forests, support vector machines, linear/quadratic discriminant analysis, neural networks (limited), deep learning (limited)
- Phylogenetics: data curation, sequence alignment, maximum likelihood, Bayesian reconstructions (limited), temporal scaling, tree shape statistics, ancestral state reconstructions (limited), discrete trait analysis (limited), phylogeography (limited), discrete trait analysis with GLMs (limited)
- Mathematical Modeling: ordinary differential equations, stochastic simulations, agent-based (limited)